

Supplementary Material

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I. DEPENDENCE OF POWER FLEXIBILITY REGION BOUNDARY IDENTIFICATION ON THE INITIAL POINT

MOST existing studies that identify the power flexibility region (PFR) boundary via discrete angular scanning start from the origin or an interior point and scan over all directions to determine boundary points by maximizing the radial distance. However, the accuracy of such algorithms can strongly depend on the selected reference point. This dependence is illustrated by the example in Fig. R1.

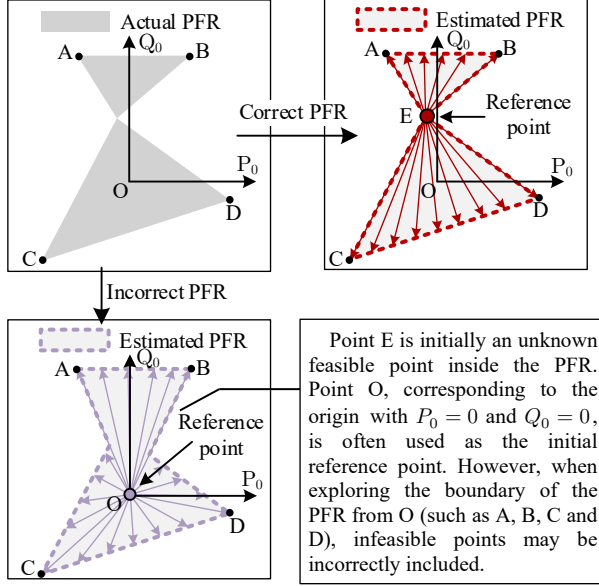


Fig. R1. Examples of PFR estimation.

In the example shown in Fig. R1, the true PFR is illustrated in Fig. R1(a). By choosing different reference points, typically the origin or a power-flow feasible point, and performing a full-angle scan, the two PFRs shown in Fig. R1(b) and (c) are obtained. It is evident that the identified PFR depends on the choice of the reference point and may be inaccurate.

II. THE PSEUDOCODE OF THE PROPOSED PFR IDENTIFICATION ALGORITHM

The complete procedure of the proposed PFR identification algorithm is given in Algorithm 1.

III. SUPPLEMENTARY DESCRIPTIONS OF THE TEST MODEL

This section provides supplementary descriptions of the relevant parameters of the modified IEEE 13-node, Swiss 38-node, and unbalanced 162-node test systems, as listed below.

A. Supplementary Description about the modified IEEE 13-node test system

Values of relevant regulating devices in the modified IEEE 13-node test system can be found in Table RI. Here, l_{min} is the gap threshold for terminating the fixed-direction interval search iteration; M is the number of scanning angles; ϵ is a numerical tolerance used in the algorithm; S_{Line} denotes the

Algorithm 1: The proposed PFR Identification Algorithm.

Input: The number of scanning angles M ; an empty feasible/infeasible set Θ_{fe}/Θ_{inf} and a temporary feasible/infeasible set $\Theta_{fe}^m/\Theta_{inf}^m$; the gap for terminating the iteration l_{min} .
Output: The feasible set Θ_{fe} and the infeasible set Θ_{inf} .

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1 for  $m = 0 : M$  do // This step can be executed in parallel.
2   Feasibility Initialization;
3   if the problem is infeasible, then
4     go to 22;
5   else
6     if  $l_{m,B} - l_{m,A} \leq l_{tol}$ , then
7        $\Theta_{inf}^m \leftarrow \Theta_{inf}^m \cup \{X_A, X_B\}$  and break;
8     else
9       Feasibility Expansion;
10      if  $l_{m,C} \geq l_{m,D}$ , then
11        go to 22;
12      else
13         $\Theta_{inf}^m \leftarrow \Theta_{inf}^m \cup \{X_C, X_D\}$ ; Feasibility Exploration;
14        if  $l_{m,E} = l_{m,C} + \epsilon$ , then
15           $\Theta_{inf}^m \leftarrow \Theta_{inf}^m \setminus \{X_C\}$ ; go to 25;
16        else
17           $\Theta_{inf}^m \leftarrow \Theta_{inf}^m \cup \{X_E, X_F\}$ ; go to 25;
18        end
19      end
20    end
21  end
22   $\Theta_{fe} \leftarrow \Theta_{fe} \cup \Theta_{fe}^m$ ;  $\Theta_{inf} \leftarrow \Theta_{inf} \cup \Theta_{inf}^m$ ;
23   $m \leftarrow m + 1$ ;
24 end
25 if  $l_{m,D} - l_{m,F} > l_{min}$ , then
26    $l_{m,C} \leftarrow l_{m,F}$ ; go to 13; //Set  $l_{m,C} \leftarrow l_{m,F}$ , and then repeat Step 3.
27 elseif  $l_{m,D} \leq l_{m,F}$ , then
28    $\Theta_{inf}^m \leftarrow \Theta_{inf}^m \setminus \{X_F\}$ ; go to 22;
29 else
30   go to 22;
31 end

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TABLE RI

VALUES OF RELEVANT PARAMETERS FOR 13-NODE SYSTEM (UNIT: P.U.)

Name	Values	Name	Values	Name	Values
l_{min}	0.005	v_{MV}^{min}	0.95	S_{PV}^{max}	1
M	120	v_{MV}^{max}	1.05	P_{load}	0
ϵ	0.001	P_{PV}^{max}	1	Q_{load}	0
S_{Line}	1	P_{PV}^{min}	0.2	$q_{C1/2/3/4}$	0/0.5/1/1.5
v_{OLTC}	0.98/1/1.02	$\tan \phi_{PV}$	1	S_c	1

TABLE RII

VALUES OF RELEVANT PARAMETERS FOR 38-NODE SYSTEM (UNIT: P.U.)

Name	Values	Name	Values	Name	Values
l_{min}	0.005	v_{MV}^{min}	0.95	S_{PV}^{max}	0.8
M	120	v_{MV}^{max}	1.05	$q_{RC1/2/3/4/5}$	-1/-0.5/0/0.5/1
ϵ	0.001	P_{PV}^{max}	0.8	S_c	1
S_{Line}	1.5	P_{PV}^{min}	0		
v_{OLTC}	0.98/1/1.02	$\tan \phi_{PV}$	1		

TABLE RIII

VALUES OF RELEVANT PARAMETERS FOR 162-NODE SYSTEM (UNIT: P.U.)

Name	Values	Name	Values	Name	Values
l_{min}	0.005	v_{MV}^{min}	0.95	S_{PV}^{max}	1
M	120	v_{MV}^{max}	1.05	S_{Line}	1
ϵ	0.001	P_{PV}^{max}	1	S_c	1
$q_{RC1/2/3/4/5}$	-1/-0.5/0/0.5/1	P_{PV}^{min}	0		
v_{OLTC}	0.98/0.99/1/1.01/1.02	$\tan \phi_{PV}$	1		

per-unit maximum line capacity; v_{OLTC} denotes the set of voltage tap positions of the on-load tap changer; $v_{MV}^{min}/v_{MV}^{max}$

are the lower and upper nodal voltage limits, respectively; P_{PV}^{max} and P_{PV}^{min} are the upper and lower active-power limits of photovoltaic (PV) units, respectively; $\tan \phi_{PV}$ denotes the maximum tangent of the PV power-factor angle; S_{PV}^{max} denotes the maximum apparent-power capacity of distributed PV; P_{load} and Q_{load} are the active and reactive loads at each node; $q_{C1/2/3/4}$ represents the per-unit reactive-power compensation corresponding to different switching levels of capacitor banks; and S_c denotes the per-unit transformer capacity.

The line parameters of the modified IEEE 13-node test system used for testing can be found in [1].

B. Supplementary Description about the Swiss 38-node test systems and unbalanced 162-node test systems

The diagrams of the Swiss 38-node test system and unbalanced 162-node test system are shown in Fig. R2 and the values of relevant regulating devices in the two test systems can be found in Table RII and RIII.

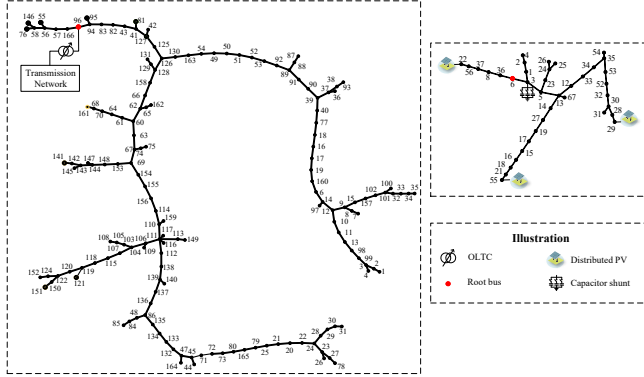


Fig. R2. Diagrams of (a) the modified 162-node test system and (b) the Swiss 38-node test system.

In Table RIII, $q_{RC1/2/3/4/5}$ denotes the reactive-power output levels of the reactor/capacitor bank. In addition, to better illustrate the nonconvexity of the PFR, the load level of the 38-node system is scaled by a factor of five.

The line parameters and load data of the 38-node Swiss test system and 162-node test system can be found in [2].

IV. THE PFR OF THE UNBALANCED MODIFIED SWISS 162-NODE TEST SYSTEM

The proposed method is further evaluated on the unbalanced modified Swiss 162-node test system. For comparison, Fig. R3 reports the estimated PFR obtained under the identical-three-phase assumption.

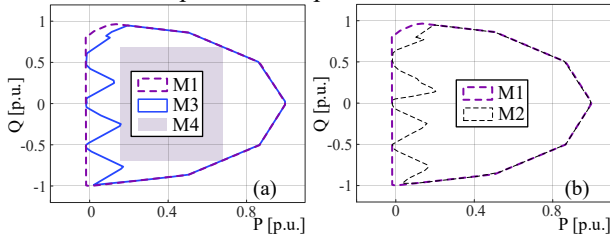


Fig. R3. PFRs obtained by different methods: (a) M1, M3, and M4 for the 162-node system; (b) M1 and M2 for the 162-node system.

V. UNCERTAINTY-CONSTRAINTS REFORMULATION AND APPLICABILITY

When the uncertainty of distributed energy resource outputs is considered, the proposed PFR identification framework can be retained if the uncertainty-related constraints are reformulated into continuous convex constraints or mixed-integer convex constraints, for every fixed binary configuration.

Consider the generic uncertain constraint as follows:

$$f(\hat{P}_{DER}, \xi) \leq 0, \quad (R1)$$

where ξ collects uncertain injections, $\hat{P}_{DER}$ is the active output of distributed energy resources.

When stochastic optimization is used to address uncertainty [3], N_s samples $\{\xi^{(n)}\}_{n=1}^{N_s}$, can be used to replace the uncertain model with a deterministic scenario-based approximation. Specifically, an uncertain constraint can be replaced by a finite family of scenario constraints:

$$f(\hat{P}_{DER}, \xi^{(n)}) \leq 0, n = 1, \dots, N_s. \quad (R2)$$

The reformulated constraints are linear and therefore remain convex.

If robust optimization is used [4], an uncertainty set $\mathcal{U}$ is constructed and feasibility is enforced for its worst realization:

$$\max_{\xi \in \mathcal{U}} f(\hat{P}_{DER}, \xi) \leq 0. \quad (R3)$$

Through robust-counterpart reformulation, constraints involving uncertain parameters can be transformed into deterministic constraints. If the original constraints are convex in the continuous variables, either unconditionally or under each fixed binary configuration, the robust-counterpart reformulation preserves the corresponding convexity.

For distributionally robust optimization [5], an ambiguity set $\mathcal{P}$ of probability distributions is used, and a representative joint chance constraint is

$$\sup_{\xi \in \mathcal{P}} \mathbb{P}\{f(\hat{P}_{DER}, \xi) > 0\} \leq \alpha. \quad (R4)$$

The original constraints involving random variables are reformulated as a finite set of mixed-integer linear constraints and are therefore convex when the binary variables are fixed.

The aforementioned distributed energy resource constraints after reformulation are convex, either directly or after fixing the binary variables. Therefore, the overall model in this letter is convex for any fixed binary-variable configuration, and the proposed method remains applicable.

VI. COMPLETE OPTIMIZATION MODEL IN THE LETTER

A. Complete Constraints of (1) in the letter

This section expands the constraints of (1) in the letter. The model includes both device and network constraints, all presented in single-period, three-phase balanced form. Further details, including the corresponding three-phase formulations, can be found in [6].

1) *Constraints of the Distributed PV:* Distributed PV units are constrained by output constraint, power factor constraint and capacity constraint (R5):

$$\begin{aligned} 0 &\leq P_{PV,i} \leq P_{PV,i}^{max}, -\tan \theta_{PV,i} P_{PV,i} \leq Q_{PV,i} \leq \tan \theta_{PV,i} P_{PV,i}, \\ \alpha_k P_{PV,i} + \beta_k Q_{PV,i} + \delta_k \cdot S_{PV,i}^{max} &\leq 0, k = 1, 2, \dots, 12, \end{aligned} \quad (R5)$$

where $\alpha_k, \beta_k, \delta_k$ are linearization coefficients.

2) *Constraints of the Capacitor Shunt*: The output $Q_{C,i}$ for a capacitor bank with K possible values $q_{C1}, q_{C2}, \dots, q_{CK}$ can be modeled and linearized using McCormick envelope relaxation (R6):

$$\begin{cases} Q_{C,i} = \sum_K q_{Ck} \lambda_{c,i,k}, \\ \begin{cases} U_i - U_i^{max}(1 - z_{c,i,k}) \leq \lambda_{c,i,k}, k = 1, 2, \dots, K, \\ U_i - U_i^{min}(1 - z_{c,i,k}) \geq \lambda_{c,i,k}, k = 1, 2, \dots, K, \\ U_i^{min} z_{c,i,k} \leq \lambda_{c,i,k} \leq U_i^{max} z_{c,i,k}, \end{cases} \\ \sum_K z_{c,i,k} \leq 1, z_{c,i,k} \in \{0, 1\}. \end{cases} \quad (R6)$$

where $\lambda_{c,i,k}$ and $z_{c,i,k}$ are continuous and binary variables representing switching states. $z_{c,i,k} = 1$ means the capacitor shunt is at step k , and the output of capacitor shunt is $q_{Ck}U_i$ and $U_i = v_i^2$.

3) *Constraints of the ESS*: The constraints of the ESS for a single period usually include, power output constraints, and complementarity constraint, which can be written as follows:

$$0 \leq P_i^{ch} \leq (1-b)P_{max}^{ch}, 0 \leq bP_i^{dis} \leq P_{max}^{dis}, b \in \{0, 1\}. \quad (R7)$$

P_i^{ch} and P_i^{dis} denote the charging and discharging powers respectively. And $P_{max}^{ch}, P_{max}^{dis}$ denote the maximum charging and discharging powers of the energy storage system. b denotes the binary variable handling the complementary constraint.

4) *Constraints of the OLTC*: The OLTC is assumed to connect the upstream grid to the root bus of the test system in this letter. The constraint regarding the ratio of the OLTC is:

$$(a_{OLTC}^{min})^2 \leq a_{OLTC} \leq (a_{OLTC}^{max})^2. \quad (R8)$$

5) *Constraints of the network*: The power-flow equations used in this paper are data-driven-aided linear power-flow model [7]:

$$\begin{aligned} v_L &= w + Y_{LL}^{-1}(\mu \text{diag}(\bar{v}_{lgt}) + (I - \mu) \text{diag}(\bar{v}_{lgt})), \\ w &= -\alpha_{OLTC} Y_{LL}^{-1} Y_{L0} v_0, \\ U_i &= \text{Re}(v_L) v_{re} + \text{Im}(v_L) v_{im}. \end{aligned} \quad (R9)$$

The remaining network constraints are as follows:

$$\begin{aligned} p_i &= \sum_{k \in \mathcal{C}(i)} P_{ik} - \sum_{k \in \mathcal{P}(i)} P_{ik}, q_i = \sum_{k \in \mathcal{C}(i)} Q_{ik} - \sum_{k \in \mathcal{P}(i)} Q_{ik}, \\ \alpha_k P_{ij} + \beta_k Q_{ij} + \delta_k \cdot S_L^{max} 1 &\leq 0, k = 1, 2, \dots, 12, \\ P_{Load,i} - P_{PV,i} + b P_i^{dis} - (1-b) P_i^{ch} &= p_i, \\ Q_{Load,i} - Q_{PV,i} - Q_{C,i} &= q_i, \\ (v^{min})^2 &\leq U_i \leq (v^{max})^2. \end{aligned} \quad (R10)$$

$P_{Load,i}$ and $Q_{Load,i}$ are the active and reactive load at node i , respectively. $\mathcal{P}(i) / \mathcal{C}(i)$ are the sets of the parent/child node of node i .

After incorporating the device and network constraints, $\mathcal{F}_c(\mathbf{X}, \mathbf{Y}, \mathbf{Z}) \leq 0$ comprises the following constraints: (R5)-(R10). The above constraints are mixed-integer linear. After fixing the integer variables, the remaining constraints reduce to linear constraints in the continuous variables, yielding a convex feasible region. Therefore, the constraints satisfy the applicability conditions of the proposed method in the letter.

B. Complete optimization problem in the letter

1) *Complete optimization problem of (5) in the letter*: In the letter, optimization problem (5) is formulated as follows:

$$l_{m,A} = \min_{\substack{\mathbf{X} \in \Omega_{PFR}^1, \\ \mathbf{Y}, \mathbf{Z}, l_m}} l_m, l_{m,B} = \max_{\substack{\mathbf{X} \in \Omega_{PFR}^1, \\ \mathbf{Y}, \mathbf{Z}, l_m}} l_m. \quad (R11)$$

The complete optimization model is presented below, taking $l_{m,A}$ as an example:

$$\begin{aligned} l_{m,A} &= \min_{\mathbf{X}, \mathbf{Y}, \mathbf{Z}, l_m} l_m \\ \text{s.t. } l_m \cos \frac{m\pi}{2M} &= p_0 + S_c, l_m \sin \frac{m\pi}{2M} = q_0 + S_c, \end{aligned} \quad (R12)$$

(R5)-(R10).

The same applies to $l_{m,B}$.

2) *Complete optimization problem of (6) and (8) in the letter*: Taking $l_{m,C}$ as an example, the complete optimization model of optimization problem (6) is formulated as follows:

$$\begin{aligned} l_{m,C} &= \max_{\mathbf{X}, \mathbf{Y}, l_m} l_m \\ \text{s.t. } l_m \cos \frac{m\pi}{2M} &= p_0 + S_c, l_m \sin \frac{m\pi}{2M} = q_0 + S_c, \end{aligned} \quad (R13)$$

(R5)-(R10)| $_{Z=Z_A}$.

3) *Complete optimization problem of (7) in the letter*: The complete optimization model of optimization problem (7) is formulated as follows:

$$\begin{aligned} l_{m,E} &= \max_{\mathbf{X}, \mathbf{Y}, \mathbf{Z}, l_m} l_m \\ \text{s.t. } l_m \cos \frac{m\pi}{2M} &= p_0 + S_c, l_m \sin \frac{m\pi}{2M} = q_0 + S_c, \end{aligned} \quad (R14)$$

(R5)-(R10), $l_m \geq l_{m,C} + \epsilon$.

VII. THE INNER CONVEX APPROXIMATION

Given the compact distribution-network model, M4 relies only on that the PFR subset associated with each fixed $\mathbf{Z}$ be convex. It constructs an inner rectangle of the nonconvex PFR such that every point within the rectangle admits an implementable discrete configuration and a feasible continuous dispatch.

M4 starts from a certified small rectangle around a feasible point and expands its active- and reactive-power limits one side at a time. Following the one-dimensional expansion strategy of [8], each side is first moved to its prescribed search limit. A certified trial is accepted; otherwise, bisection determines the outermost certified boundary. Because the expansion order affects the attainable area, several representative orders are evaluated from the same initial rectangle, and the largest candidate is retained. Only certified expansions are accepted, so the incumbent rectangle remains certified throughout the search.

Each candidate is certified using a max-min test. Two outer binary variables select the worst-case corner, while the inner problem adjusts $\mathbf{Y}$ to minimize the interface-power mismatch. For a fixed $\mathbf{Z}$, if the mismatches at all four corners are within the prescribed tolerance, convexity certifies the entire rectangle. If no single configuration covers a large candidate, M4 adaptively partitions it into leaf cells and assigns a separate $\mathbf{Z}$ to each cell. The candidate is certified only if all leaf cells pass the corner tests and cover it without gaps. A failed test indicates only that the candidate cannot be certified, rather than that it is physically infeasible.

Finally, M4 independently re-evaluates every leaf cell of the largest candidate using the max-min test. The rectangle is returned only if all worst-case mismatches satisfy the safety tolerance. If the search times out, M4 returns the largest previously certified rectangle.

The search procedure of M4 is illustrated in Fig. R4.

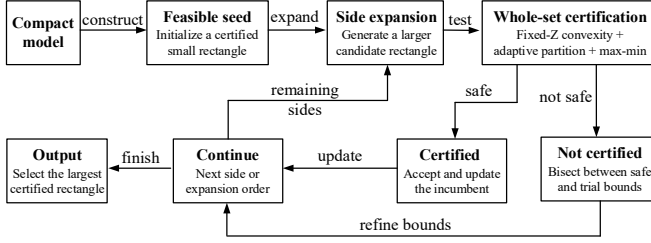


Fig. R4. M4 search procedure for a certified inner rectangle.

VIII. CORRESPONDENCES BETWEEN SOME COLORED REGIONS AND THE BINARY VARIABLES

This section uses the modified IEEE 13-node test system as an example. It should be noted that although the case study considers a balanced three-phase system for the modified IEEE 13-node test system, the OLTC tap positions and capacitor-bank switching states are modeled as independently adjustable across phases. Since one colored region may correspond to multiple binary-variable configurations, it is not possible to enumerate all cases. Therefore, only several representative correspondences are presented in Fig. R5 and Table RIV.

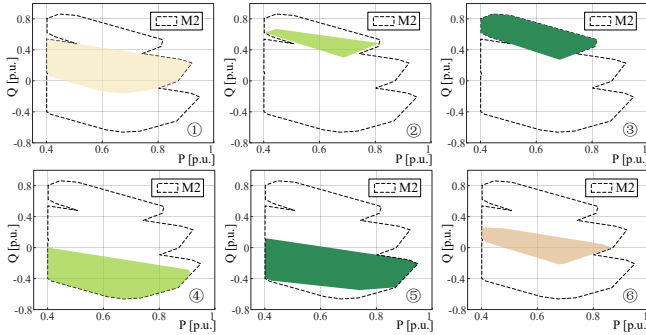


Fig. R5. Colored regions in the IEEE 13-node test system.

TABLE RIV
BINARY CONFIGURATIONS OF COLORED REGIONS

Region label	OLTC tap position	Capacitor bank position (phase a/b/c)
①	1.00/0.98/0.98	0/1/1
②	1.00/0.98/0.98	0/0/0
③	1.02/0.98/0.98	0/0.5/0
④	1.00/1.00/1.00	0/1/0
⑤	0.98/0.98/0.98	0/0/0
⑥	0.98/0.98/0.98	0.5/0/0

IX. SUPPLEMENTARY DESCRIPTION OF THE PERFORMANCE METRICS IN TABLE I

The intersection-over-union (IoU) is defined as the ratio of the overlap area between two regions to their total covered area. The infeasible inclusion ratio (IIR) and the feasible omission ratio (FOR) represent the proportions of infeasible operating points incorrectly included and feasible operating points incorrectly excluded, respectively.

Let A denote the estimated region obtained by a candidate

method and B denote the benchmark region obtained by the enumeration method, and let $|\cdot|$ denote the area of a region. Then, the IoU, IIR and FOR are defined as (1)-(3):

$$IoU = \frac{|A \cap B|}{|A \cup B|}, \quad (R15)$$

$$IIR = \frac{|A \setminus B|}{|B|}, \quad (R16)$$

$$FOR = \frac{|B \setminus A|}{|B|}. \quad (R17)$$

A larger IoU indicates higher geometric similarity. Smaller IIR and FOR values indicate fewer incorrectly included infeasible operating points and fewer incorrectly omitted feasible operating points, respectively.

X. SUPPLEMENTARY DESCRIPTION OF THE PROPOSED PFR IDENTIFICATION ALGORITHM

A. Proof of Convergence of the PFR Identification Algorithm

For each scanning direction, the fixed-direction interval search procedure explores binary configurations along the corresponding ray. In Step 1, two boundary configurations are obtained by solving the minimum- and maximum-distance problems. If these two configurations do not provide a continuous feasible segment, Step 3 searches for the next feasible segment by imposing $l_m \geq l_{m,C} + \epsilon$. This constraint prevents the algorithm from repeatedly returning the same feasible point within the prescribed tolerance. Since the number of binary configurations is finite, the number of newly explored configurations along a ray is also finite. Therefore, the fixed-direction interval search procedure terminates after a finite number of iterations. Let N_{01} denote the total number of binary configurations. The worst-case number of fixed-direction interval search iterations for each scanning direction is bounded by N_{01} , and the overall worst-case number of explored configurations is bounded by $O(MN_{01})$. In summary, the algorithm is guaranteed to converge within a finite number of iterations.

B. Sensitivity of the proposed algorithm to M , l_{min} and l_{tol}

In this section, the sensitivity of the proposed PFR identification algorithm to the number of angular partitions M , the gap for terminating the fixed-direction interval search iteration l_{min} , and the small positive tolerance ϵ is evaluated on the 13-node test system. In the following tables, the IoU always denotes the intersection-over-union between the PFR obtained by the proposed method and that obtained by the enumeration method. The results are summarized as follows.

We first assess the sensitivity of the proposed algorithm to M . The test scenario is identical to the 13-node test system presented in the manuscript, except that only the value of M is changed.

TABLE RV
SENSITIVITY OF THE PFR IDENTIFICATION ALGORITHM TO M

M	15	30	60	120	240	360
IoU	0.875	0.955	0.993	0.995	0.995	0.995

It can be observed that as M increases, the IoU first increases and then tends to remain unchanged. This indicates that the estimation accuracy of the PFR generally improves with the

increase of M . However, beyond a certain threshold, for example when $M = 120$, the improvement in PFR accuracy becomes marginal and the IoU remains almost unchanged. Therefore, in practical applications, M should be properly selected to achieve a trade-off between evaluation accuracy and computational cost.

Then, the sensitivity of the proposed algorithm to l_{min} and ϵ is evaluated. Except for the parameter under evaluation, all other settings are the same as those in the manuscript.

TABLE RVI

SENSITIVITY OF THE PFR IDENTIFICATION ALGORITHM TO l_{min} AND l_{tol}		
l_{min}	ϵ	IoU
0.005	0.0001	0.995
	0.001	0.995
	0.01	0.995
	0.1	0.995
0.001	0.001	0.995
0.005		0.995
0.05		0.979
0.5		0.956

It can be observed that, in the test case, increasing ϵ has almost no impact on the IoU. This is because ϵ mainly affects the identification of interior infeasible points, whereas the tested case contains no interior infeasible points and only exhibits nonconvexity near the boundary. Therefore, the resulting impact is limited. In practical tests, it is still recommended that ϵ be set smaller than l_{min} . In addition, as l_{min} further increases, the IoU generally decreases. This is because l_{min} affects the identification of nonconvex boundaries and interior infeasible regions of the PFR. An excessively large l_{min} may cause feasible points to be incorrectly classified as infeasible during the PFR identification process, thereby reducing the IoU. Therefore, in the test case, a relatively small value of l_{min} , such as 0.005, is recommended.

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